Hands-on: Three Operator Cases

Help an operator compare options, check results and decide the next step.

One notebook / 90 minutes / three short exercises

Alaa Selim
MIT/Dartmouth

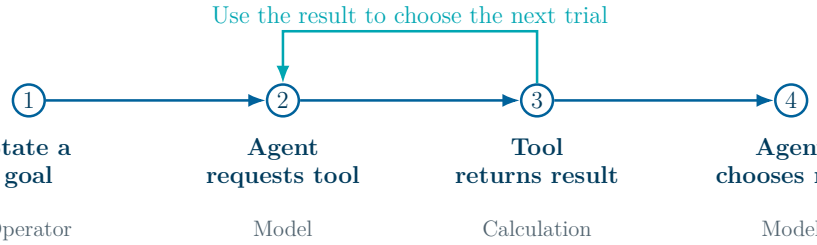
Mohamed Chehada
UT Austin



Independent educational material / IEEE ComSoc theme adaptation

Source: Setup: COLAB_GUIDE.md; Classroom_Labs.ipynb.

An agent chooses what to try next



Teaching diagram: the model requests a tool and uses the returned evidence.

- Give the agent a goal. It requests a tool, reads the result and chooses the next step.

Source: Monica Lam. CS224V Agentic AI, Lecture 1: Introduction. Stanford University, Fall 2026, p.2.

The agent asks; the tool calculates



Original Stanford figure: one function call and a multi-step workflow.

- A tool does one job. An agent can connect several jobs using their results.

Source: Monica Lam. CS224V Agentic AI, Lecture 1: Introduction. Stanford University, Fall 2026, p.2.

Check the evidence before answering



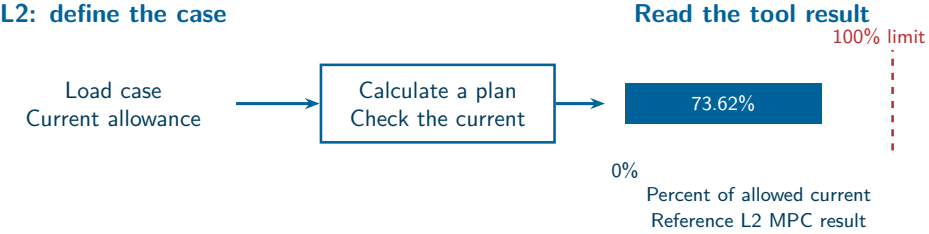
Grid teaching adaptation of Stanford's idea: check claims before reporting.

- Keep the tool result and its checks with the recommendation.

Source: Monica Lam. CS224V Agentic AI, Lecture 1: Introduction. Stanford University, Fall 2026, p.19.

Solar settings: which option should we try?

Colab / L2



Teaching adaptation. L2 runs Python; L10B replays saved model choices.

- Compare possible settings, then check voltage and line current.
- The agent can organize these trials and report the checked options.

Source: Monica Lam. CS224V Agentic AI, Lecture 1: Introduction. Stanford University, Fall 2026, p.19; Classroom Labs L2.

Give the agent a goal and clear limits



State the goal

What should be achieved?

Name allowed tools

What may be requested?

Set stop rules

When is review needed?

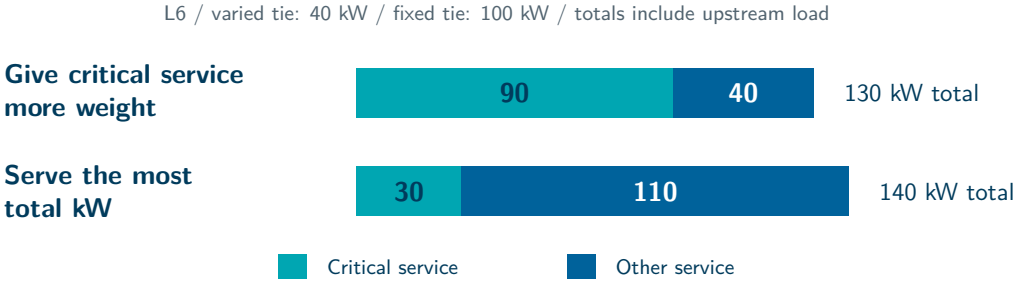
Grid teaching adaptation of Stanford's task-specification concept.

- State the goal, allowed tools and stopping rules.
- Keep the operator's review step explicit.

Source: Monica Lam. CS224V Agentic AI, Lecture 1: Introduction. Stanford University, Fall 2026, p.40.

Restoration: who should get power first?

Colab / L6



L6: same network and limits, different service goals.

- The operator sets the service priority.
- That priority changes what we ask the restoration tool to do.

Source: Monica Lam. CS224V Agentic AI, Lecture 1: Introduction. Stanford University, Fall 2026, p.40; Classroom Labs L6.

Keep three jobs clear

Stanford teaching idea / grid adaptation



1	2	3
Agent requests	**Tool calculates**	**Operator decides**
Choose the study	Return the result	Review the next step

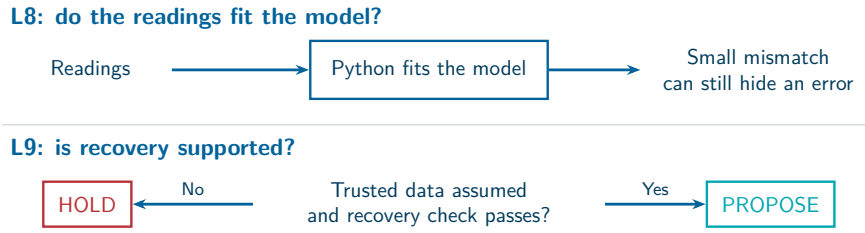
Adapted principle: language, calculation and human review have different jobs.

- The agent requests a study. The tool calculates. The operator reviews the result.

Source: Monica Lam. CS224V Agentic AI, Lecture 1: Introduction. Stanford University, Fall 2026, p.53.

Wrong readings can still fit a model

Colab / L8–L9



L9 assumes those extra readings are trusted. The numerical tool does not authenticate them.

L8 checks fit. L9 requires a trust assumption before proposing recovery.

- A good fit alone does not prove the readings are true.
- The agent can organize checks and flag missing evidence.

Source: Monica Lam. CS224V Agentic AI, Lecture 1: Introduction. Stanford University, Fall 2026, p.53; Classroom Labs L8–L9.

Ask four questions about every result

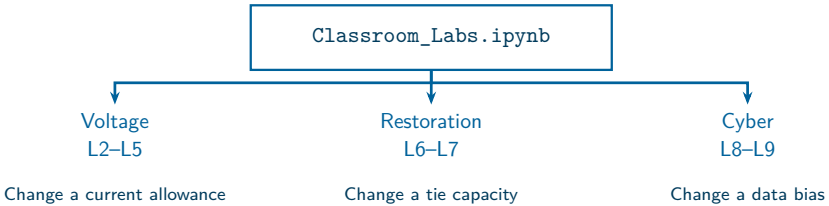
1. **Evidence**
   Which tool result supports the claim?
2. **Assumptions**
   Which inputs and model were used?
3. **Rules**
   Were the same limits and priorities kept?
4. **Next step**
   What does the operator still need to review?

Teaching adaptation of the cited lecture. The lab uses custom Python.

- What was asked? What did the tool return?
- Did the checks pass? What must the operator decide?

Source: Monica Lam. CS224V Agentic AI, Lecture 1: Introduction. Stanford University, Fall 2026, pp.52–53.

Use one notebook for all three cases



L10B replays Llama choices in the same file. Each case uses a separate teaching network.

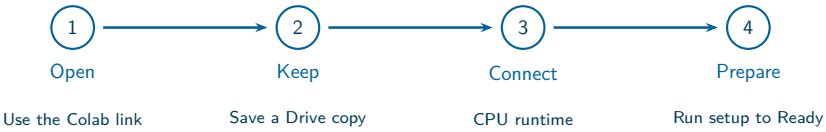
Classroom Labs is the only required notebook.

Open Classroom Labs in Google Colab

- Open Classroom Labs and save your own copy.
- We will use it throughout the workshop.

Source: Setup: COLAB_GUIDE.md; Classroom_Labs.ipynb.

Start Colab in three steps



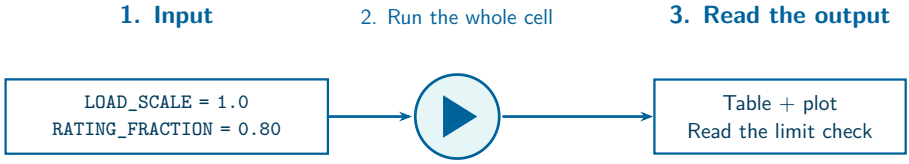
For the guided workshop, run each named lesson when prompted. Run all is a useful setup check.

Setup first, then L0 and L1.

- Save a copy, connect a CPU and run setup.
- Wait for Ready, then run L0 and L1.

Source: Setup: COLAB_GUIDE.md; Classroom_Labs.ipynb.

Run a cell, then read its output



The play button executes Python. The saved text below a cell is its output.

Illustration of a code cell. Run the full L2 cell.

- Find the L heading and run the whole cell.
- After changing an input, run that cell again.

Source: Google Colab FAQ; Classroom Labs L2, cell classroom-007.

L0: keep both switches off

Colab / L0

Colab L0 / run the whole cell

```
RUN_OPENDSS = False       # True: fresh full-model audit +
   separate AC MPC solve in L5
RUN_LOCAL_LLAMA = False   # True: one optional local call in L11
```

- Keep both switches False. Python runs the calculations.
- Llama, the language model, uses saved replies here. No live model is needed.

Source: Classroom_Labs.ipynb, L0–L1; cells classroom-002 and classroom-004.

L1: does the old result still match?

Colab / L1

Executed notebook output

L1 · Does the evidence match?

classroom-004 · output 0 (zero-based) · recorded execution 3

	test	binding_pass
0	same request + same command	True
1	changed load; old evidence	False
2	changed command; old evidence	False

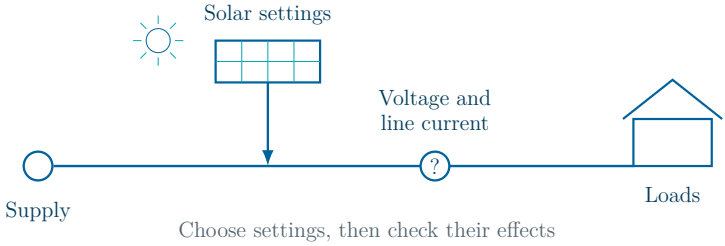
Reference L1 output. Only the unchanged request and command match.

- Expect True, False, False: only the unchanged request matches.
- Changed inputs or actions need a new check.

Source: Classroom_Labs.ipynb, L0–L1; cells classroom-002 and classroom-004.

Case 1: which solar settings should we use?

Case 1 / The operator's problem

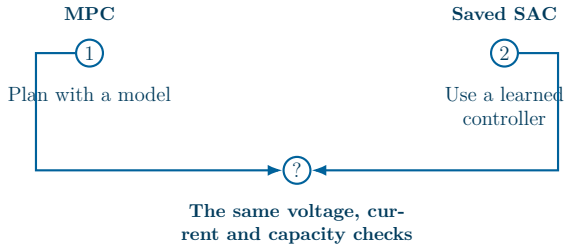


Concept sketch. First run the tools; later replay saved agent decisions.

- Choose settings that keep voltage and line current within the allowed range.
- An agent can organize the trials and return checked options.

Source: Classroom_Labs.ipynb, L2; cell classroom-007; src/continuous_control.py.

Two tools suggest different settings



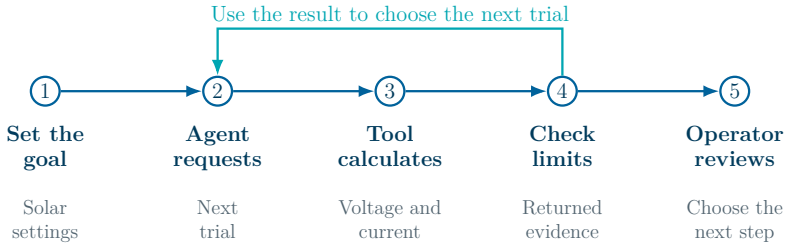
MPC: model predictive control. SAC: soft actor-critic. Both proposals need the same checks.

- MPC plans with a model. SAC uses a previously trained controller.
- Treat both outputs as proposals that need the same electrical checks.

Source: Classroom Labs notebook. Classroom_Labs.ipynb, L2; cell classroom-007; src/continuous_control.py. Public notebook and Python source. 24 October 2026; stable notebook cell IDs checked.

How the agent helps with solar settings

How an agent can help



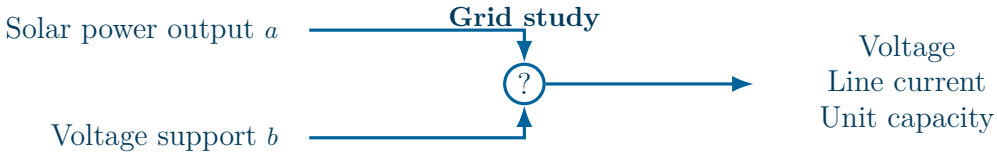
Concept map. Colab runs the tools; L10B replays the agent decisions.

- The agent requests a study and reads the returned checks.
- It can try another allowed option or ask the operator for help.

Source: Workshop recorded model replies and numerical study records. Voltage P3_current_reserve: Llama 3.1:8b planner, recovery and review calls 1–3. Public teaching archive. 27 September 2026.

The study changes two solar settings

Colab / L2

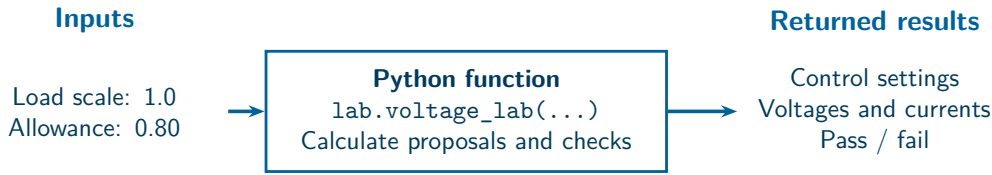


Teaching view of the model's two controls. Inverter capacity is also a limit.

- a sets active-power output. b sets voltage-support output.
- The tool checks voltage, line current and the solar unit's capacity.

Source: Classroom_Labs.ipynb, L2; cell classroom-007; src/continuous_control.py.

A Python function can be a tool



In L2, your cell calls this function directly.

In the recorded agent study, Python checks Llama's choice before running an allowed tool.

L2 calls lab.voltage_lab directly. Agent coordination is shown in the saved study.

- It receives inputs, runs calculations and returns results.
- L2 calls this function directly. The saved agent example shows who requests it.

Source: Classroom Labs notebook. Classroom_Labs.ipynb, L2; cell classroom-007; src/continuous_control.py. Public notebook and Python source. 24 October 2026; stable notebook cell IDs checked.

L2: compare two solar proposals

Colab / L2

Colab L2 / run the whole cell

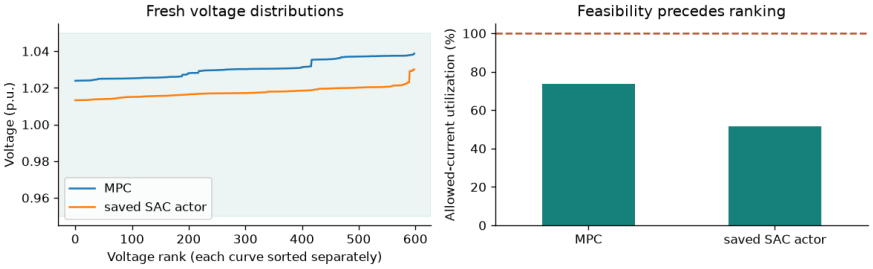
```
LOAD_SCALE = 1.0
RATING_FRACTION = 0.80
started = time.perf_counter()
voltage = lab.voltage_lab(LOAD_SCALE, RATING_FRACTION)
```

- Run L2 with the default settings.
- Look for pass/fail, current used and score J. Lower J is better after checks pass.

Source: Classroom_Labs.ipynb, L2; cell classroom-007; src/continuous_control.py.

Read the voltage and current checks

Colab / L2



Reference L2 plots. Rerun for your chosen inputs.

- Left: voltages should stay in the allowed band.
- Right: current should stay below 100% of the allowance.

Source: Classroom_Labs.ipynb, L2; cell classroom-007; src/continuous_control.py.

Both pass; MPC has the lower score

Colab / L2

Executed notebook output

L2 · Propose and check voltage settings

classroom-007 · output 1 (zero-based) · recorded execution 4

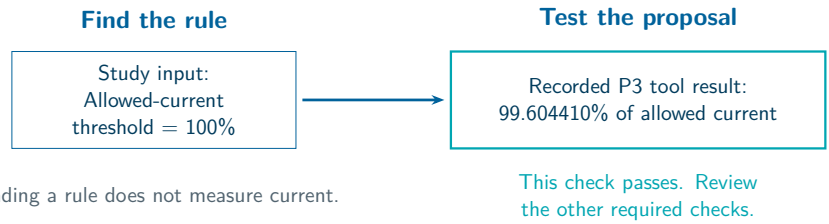
	method	feasible	current_ step_J	a	b	vmin	vmax	max_allowed_ current_pct	failures
0	MPC	True	0.031104	0.960307	-0.111114	1.023874	1.038709	73.619901	none
1	saved SAC actor	True	0.436428	0.340066	0.014024	1.013243	1.030000	51.679336	none

Reference L2 output at load scale 1.0 and current allowance 0.80.

- MPC: 73.62% current use; score J about 0.03110.
- Saved SAC: 51.68% current use; score J about 0.43643.

Source: Classroom_Labs.ipynb, L2; cell classroom-007; src/continuous_control.py.

A limit needs a calculated check



Example uses the separate saved P3 case.

- A rule states the limit. The tool tests the proposed setting.
- Keep the rule and returned result together for review.

Source: Monica Lam. CS224V Agentic AI, Lecture 1: Introduction. Stanford University, Fall 2026, pp.17–25; recorded P3 current check.

Exercise 1: lower the current allowance

Try it in Colab / L2

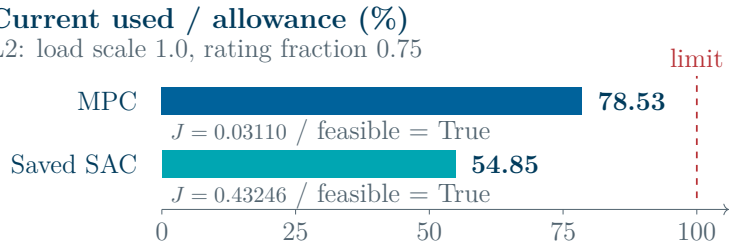
```
RATING_FRACTION = 0.75
```

- In L2, change RATING_FRACTION from 0.80 to 0.75. Predict, then rerun.
- Compare pass/fail, current use and score J for both methods.
- Does this check cover the whole medium- and low-voltage network?

Source: Classroom_Labs.ipynb, L2; cell classroom-007; src/continuous_control.py.

Exercise 1 answer: both options still pass

Worked answer / L2

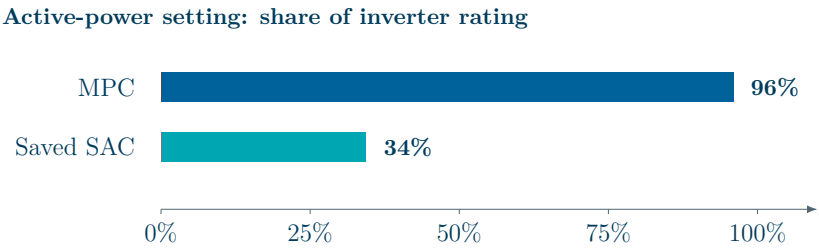


Fresh L2 exercise results. Current is shown relative to the changed allowance.

- MPC has the lower score; SAC uses less current. The score considers more than current alone.
- This check covers the medium-voltage model. The wider network needs a wider study.

Source: Classroom_Labs.ipynb, L2; cell classroom-007; fresh numerical check at rating fraction 0.75.

Less current does not mean the best option



Exercise values at RATING_FRACTION=0.75; a uses inverter rating as its base.

- Here SAC also uses less solar output.
- The goal values generation as well as limits. Compare the score after checks pass.

Source: Classroom Labs L2, cell classroom-007; verified default and exercise outputs.

L2: restore the starting values

Colab / L2

Colab L2 / run the whole cell

```
LOAD_SCALE = 1.0
RATING_FRACTION = 0.80
```

- Restore LOAD_SCALE=1.0 and RATING_FRACTION=0.80. Rerun L2.
- The next cells use a separate saved case.

Source: Classroom_Labs.ipynb, L2; cell classroom-007; src/continuous_control.py.

Separate a new calculation from a replay

A fresh calculation



A recorded replay



A replay keeps its original inputs, decisions and results.

- Your inputs produce new numerical results.
- Saved replies show past model decisions; editing L2 does not change them.

Source: Classroom Labs notebook. Setup: COLAB_GUIDE.md; Classroom_Labs.ipynb. Public notebook and Python source. 24 October 2026; stable notebook cell IDs checked.

L3: inspect saved model choices

Colab / L3

Colab L3 / run the whole cell

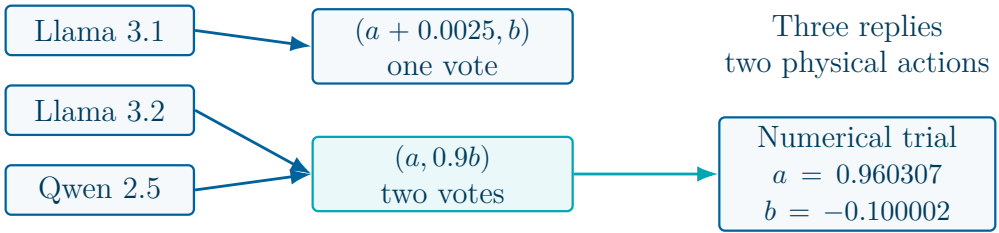
```
pool = lab.saved_pool('test_01')
```

- Run L3 to see what the models requested for the saved case.
- This displays old replies; it makes no new model call.

Source: Classroom_Labs.ipynb, L3; cell classroom-010; archived test_01 context.

Different replies can request the same action

Colab / L3



Three saved replies request two different trials.

- Group requests that describe the same settings.
- A vote chooses what to test next. The electrical check still has to run.

Source: Classroom_Labs.ipynb, L3; cell classroom-010; archived test_01 context.

L4: check the chosen proposal

Colab / L4

Colab L4 / run the whole cell

```
selected_trials = lab.trial_saved_selection(pool)
```

- Recheck the selected proposal on its original saved case.
- Compare the vote with a previously learned selection rule.

Source: Classroom_Labs.ipynb, L4; cell classroom-013; fresh checks of archived choices.

Both selectors choose the same proposal

Colab / L4

Executed notebook output

L4 · Check the selected trial

classroom-013 · output 1 (zero-based) · recorded execution 6

	selector	operation	new_check_passes	current_step_J	verification_s	failures
0	majority vote	RELAX_Q_10_PERCENT	True	0.031082	0.081254	none
1	frozen LinUCB	RELAX_Q_10_PERCENT	True	0.031082	0.111011	none

Reference L4 output: both selection methods choose the same action.

- The new electrical check passes for both selections.
- The selection score and the electrical check have different jobs.

Source: Classroom_Labs.ipynb, L4; cell classroom-013; fresh checks of archived choices.

L5: see which part of the grid was checked

Colab / L5

Colab L5 / run the whole cell

```
scope_audit = lab.backend_audit(pool, run_opendss=RUN_OPENDSS)
```

- Run L5 with the default settings.
- It checks the medium-voltage (MV) part of the saved case.

Source: Classroom_Labs.ipynb, L5; cell classroom-016; MV-only default model.

A PASS covers only what was checked

Colab / L5

Executed notebook output

L5 · What does the model cover?

classroom-016 · output 0 (zero-based) · recorded execution 7

	model	command	fresh	passes	vmin	vmax	current_pct	failures
0	lindistflow3	same archived MV-MPC first action	True	True	1.023874	1.038709	73.661582	none

Reference L5 result covers medium voltage only.

- A limited model cannot answer a wider grid question.
- Request the missing study or return HOLD.

Source: Classroom_Labs.ipynb, L5; cell classroom-016; MV-only default model.

Recorded example: the first option fails

Saved voltage example / separate from L2

RECORDED REPLAY · llama3.1:8b · voltage · call 1	
Input facts	Forecast load multipliers: [1.0, 0.94, 1.06]. Current allowance: 80% of nominal; voltage limits: [0.95, 1.05] p.u.
Llama choice	`VERIFY_SAC_FIRST`
Exact public reason	“A numerical SAC proposal is available; its fresh first-action check has not yet been requested.”
Recorded result	OpenDSS: 111.966607% of the allowed current; 2 current violations. Deterministic check: REJECT; failure = thermal.

- This saved study is different from your L2 run.
- Llama asks for a SAC check. The tool reports too much current.

Source: Classroom Labs L10B: voltage replay, recorded call 1; full record in source guide.

Recorded example: request another option

Saved voltage example / separate from L2

RECORDED REPLAY · llama3.1:8b · voltage · call 2

Input facts	The supplied failed checks are: thermal. An MPC plan already exists; its recovery-eligibility failure list is empty.
Llama choice	`USE_MPC_PLAN`
Exact public reason	“The MPC plan has been generated and is eligible for a numerical check. This is the next allowed step in the recovery process.”
Recorded result	OpenDSS: 99.604410% of allowed current; no recorded violations. Proposed P = 2387.157 kW; Q = +325.738 kvar.

- Llama asks the tool to check the existing MPC fallback.
- The returned electrical checks pass for this saved case.

Source: Classroom Labs L10B: voltage replay, recorded call 2; full record in source guide.

Recorded example: return to the operator

Saved voltage example / separate from L2

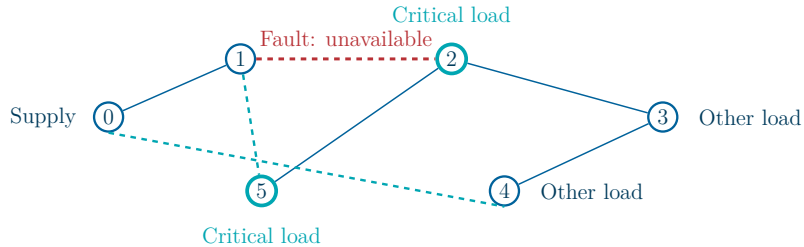
RECORDED REPLAY · llama3.1:8b · voltage · call 3	
Input facts	The full-feeder first-action check passed; voltage is 1.001783–1.035026 p.u. Protection, transients and implementation authority remain outside this model.
Llama choice	`REVIEW_EVIDENCE`
Exact public reason	“The model has converged and the numerical evidence indicates that the control actions are within the allowed limits. Reviewing the evidence will provide more insight into the model's performance and the selected control actions.”
Recorded result	Numerical status: MODEL_RECOMMEND; recorded human acknowledgment: PENDING. Acknowledged = False; dispatch authorized = False; equipment operated = False.

- The agent returns the checked proposal and its conditions.
- Human review is still pending. No equipment command is sent.

Source: Classroom Labs L10B: voltage replay, recorded call 3; full record in source guide.

Case 2: who gets power after a fault?

Case 2 / The operator's problem



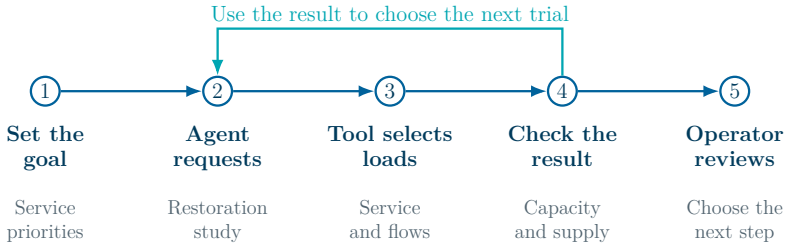
Six-node lab graph. Cyan marks critical loads; dashed links are backup options.

- A fault removes a supply path. Available capacity limits what can be restored.
- The operator sets priorities; the agent can request and check restoration options.

Source: Classroom_Labs.ipynb, L6; cell classroom-018; src/standard_formulations.py.

How the agent helps restore service

How an agent can help



Concept map. L6 runs the solver directly; L10B replays a separate saved agent study.

- Choose the goal: critical loads first, or the most total power.
- The agent can request a study, read the trade-off and ask for a new study when limits change.

Source: Classroom_Labs.ipynb, L6; cell classroom-018; src/standard_formulations.py.

L6: compare two restoration goals

Colab / L6

Colab L6 / run the whole cell

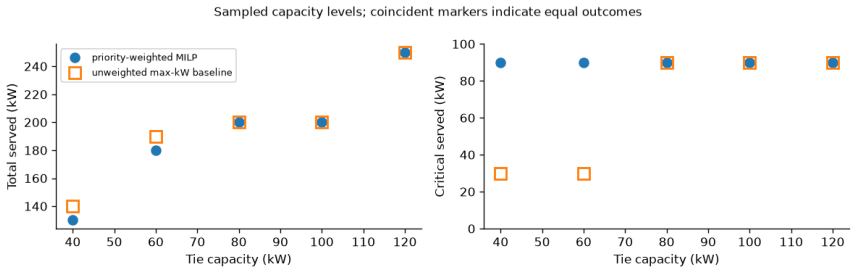
```
CAPACITIES_KW = (40., 60., 80., 100., 120.)
started = time.perf_counter()
restoration = lab.restoration_sweep(CAPACITIES_KW)
```

- Run L6 to choose which loads receive power under each goal.
- MILP chooses whole loads; a separate calculation checks its result.

Source: Classroom_Labs.ipynb, L6; cell classroom-018; src/standard_formulations.py.

Read total and critical service

Colab / L6



Reference L6 results. Both goals use the same network.

- Left: all power restored. Right: power restored to critical loads.
- Overlapping markers mean the two goals give the same result.

Source: Classroom_Labs.ipynb, L6; cell classroom-018; src/standard_formulations.py.

The service goal changes who gets power

Colab / L6

Executed notebook output

L6 · Which loads should return first?

classroom-018 · output 0 (zero-based) · recorded execution 8

Actual output header and rows 0–1 · tie_capacity_kw = 40.0

	tie_capacity_kw	method	served_kw	critical_kw	weighted_service	MILP_matches_exhaustive_check
0	40.0	priority-weighted MILP	130.0	90.0	940.0	True
1	40.0	unweighted max-kW baseline	140.0	30.0	410.0	True

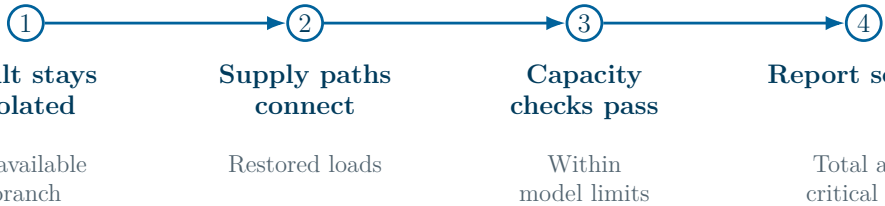
Reference L6 output at the 40 kW tie setting.

- Critical-first goal: 130 kW total, including 90 kW critical.
- Most-total-power goal: 140 kW total, including 30 kW critical.

Source: Classroom_Labs.ipynb, L6; cell classroom-018; src/standard_formulations.py.

Check the restoration plan

Colab / L6



Checks on the active-power model. AC and field-operation checks remain separate.

- Keep the faulted branch out of service.
- Check supply paths, capacity and totals before accepting the plan.

Source: Classroom_Labs.ipynb, L6; cell classroom-018; src/standard_formulations.py.

Exercise 2: increase tie capacity

Try it in Colab / L6

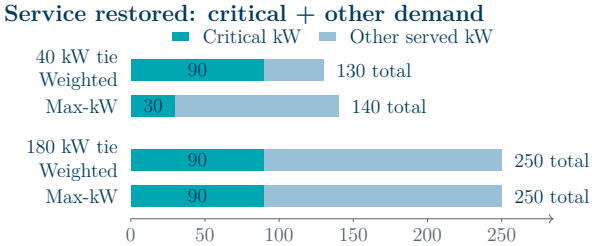
```
CAPACITIES_KW = (40., 60., 80., 100., 120., 180.)
```

- Add 180. to CAPACITIES_KW. Keep 40. first. Predict, then rerun L6.
- Compare total and critical power at 40 and 180 kW.
- At which tested capacity do the two goals first give the same result?

Source: Classroom_Labs.ipynb, L6; cell classroom-018; src/standard_formulations.py.

Exercise 2 answer: capacity changes the trade-off

Worked answer / L6



Fresh L6 results. Cyan shows critical service; blue shows other service.

- At 40 kW, the priority goal serves 90 kW critical instead of 30 kW, with less total service.
- The goals first match at the tested 80 kW. At 180 kW, both serve 250 kW total and 90 kW critical.

Source: Classroom_Labs.ipynb, L6; cell classroom-018; fresh MILP and exhaustive-check sweep.

Colab / L6

250 kW total / 90 kW critical

- At 180 kW, both goals give the same result.
- The chosen priority matters when capacity forces a choice.

Source: Classroom_Labs.ipynb, L6; cell classroom-018; src/standard_formulations.py.

L7: change the request, then check the evidence

Colab / L7

Colab L7 / run the whole cell

```
print('Original contract:', lab.bound_to(rest_request,
    rest_candidate, rest_evidence))
print('Changed rating with old evidence:', lab.bound_to(
    new_rest_request, rest_candidate, rest_evidence))
```

- Restore the original capacity list and rerun L6.
- Run L7: the requested capacity changes from 40 kW to 20 kW.

Source: Classroom_Labs.ipynb, L7; cell classroom-021; request/evidence binding.

The old result no longer matches

Colab / L7

Executed notebook output

L7 · Changed capacity needs a new check

classroom-021 · output 0 (zero-based) · recorded execution 9

```
Original contract: True
Changed rating with old evidence: False
Required response: HOLD old recommendation; request a fresh constrained solve.
```

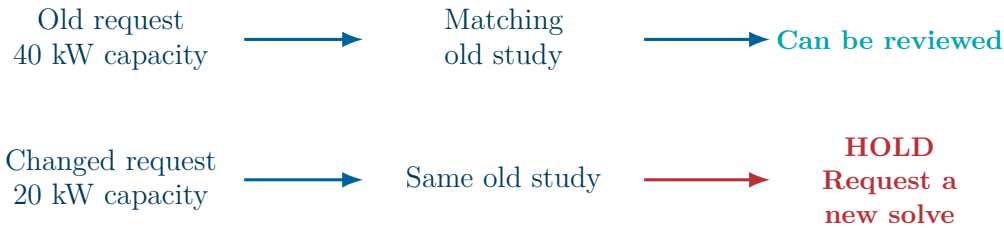
Reference L7 result: changed inputs no longer match the old evidence.

- The original request matches; the changed request does not.
- Put the old recommendation on HOLD and request a new calculation.

Source: Classroom_Labs.ipynb, L7; cell classroom-021; request/evidence binding.

A new limit needs a new study

Colab / L7



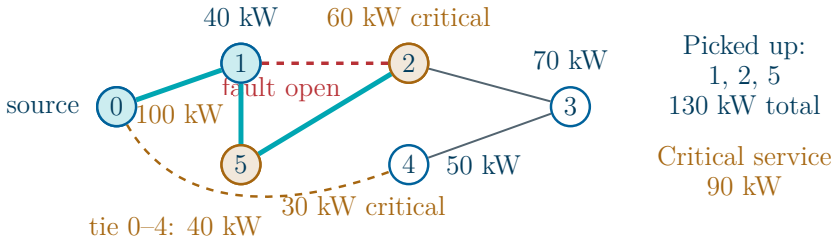
A failed input match asks for a new study; it does not establish physical infeasibility.

- Old evidence does not tell us whether a new plan works.
- Calculate again, check again and attach the new result to the new request.

Source: Classroom_Labs.ipynb, L7; cell classroom-021; request/evidence binding.

The agent keeps the operator's goal

Colab / L6



Proposed agent use. L6 itself runs Python.

- The operator sets priorities. The solver selects loads.
- The agent can request the solve and explain the returned service trade-off.

Source: Classroom_Labs.ipynb, L6; cell classroom-018; src/standard_formulations.py.

This model leaves some checks for later

Colab / L6

- This small model checks which loads receive power.
- Voltage, protection and field switching need separate checks.

Source: Classroom_Labs.ipynb, L6; cell classroom-018; src/standard_formulations.py.

Recorded example: try more service

Saved feeder example / separate from L6

RECORDED REPLAY · llama3.1:8b · restoration · call 14	
Input facts	E01 failed its thermal check; E02 passed. E02 serves 1242.777 kW with 65.938897% tie loading.
Llama choice	SHED_075
Exact public reason	“The candidate E01 has a thermal failure, and the lowest cost passed candidate E02 has a cost of -5509.135240000002. Selecting SHED_075 will test 75% disconnection of the ordered noncritical load objects, which may improve the cost.”
Recorded result	SHED_075 produces E03: 1843.614 kW served and 474.040 critical kW. E03 passes at 91.867215% tie loading; cost = -6109.971300 (smaller is better).

- This saved feeder study is different from L6.
- Llama requests another plan. The tool reports more service and passing checks.

Source: Classroom Labs L10B: restoration replay, recorded call 14; full record in source guide.

Recorded example: the tool rejects a trial

Saved feeder example / separate from L6

RECORDED REPLAY · llama3.1:8b · restoration · call 15	
Input facts	E03 is the lowest-cost passed candidate; one refinement evaluation remains. SHED_065 means disconnecting 65% of ordered noncritical load objects, not 65% of kW.
Llama choice	`SHED_065`
Exact public reason	“The candidate E01 has a thermal failure, and the lowest cost passed candidate E03 has a cost of -6109.9713, which is lower than the objective's target. Selecting SHED_065 will test 65% disconnection of the ordered noncritical load objects, which may improve the cost.”
Recorded result	E04 reaches 111.009413% tie loading: REJECT, thermal failure. The rejected trial has no eligible score; its larger served load cannot make it acceptable.

- The next trial carries too much current.
- It cannot replace the previous passing option.

Source: Classroom Labs L10B: restoration replay, recorded call 15; full record in source guide.

Recorded example: correct a wrong ranking

Saved feeder example / separate from L6

RECORDED REPLAY · llama3.1:8b · restoration · call 16

Input facts	E02 cost = -5509.135240; E03 cost = -6109.971300; smaller is better. Only the passed candidates E03 and E02 are offered for ranking.
Llama choice	`E02, E03`
Exact public reason	“The candidates were ranked based on their scores, with the lowest score (best) first. E02 has a score of -5509.135240000002, and E03 has a score of -6109.971300000003.”
Recorded result	Returned language order: E02 → E03. Correct numerical order: E03 → E02. The guard retains E03 and rechecks it as E05: 1843.614 kW served; no recorded failures.

- Llama puts two passing plans in the wrong order.
- Python keeps the lower-cost plan and checks it again.

Source: Classroom Labs L10B: restoration replay, recorded call 16; full record in source guide.

Case 3: can we trust these readings?

Case 3 / The operator's problem

Some readings may be wrong



A good fit is not the same as true data

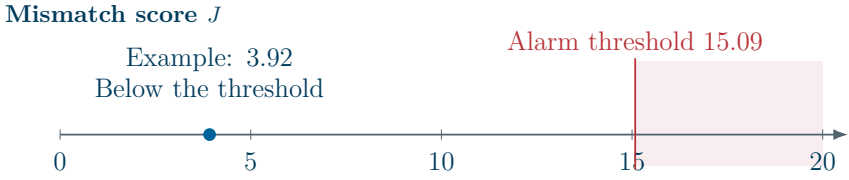
Concept sketch. L8 uses a small four-bus DC model.

- Wrong readings can still look consistent with the model.
- The agent can request checks, but action still needs a trusted basis.

Source: Classroom_Labs.ipynb, L8; cell classroom-023; four-bus DC fixture.

Mismatch is only one check

Colab / L8



A low mismatch does not authenticate a reading

Mismatch J here has a different meaning from control cost J in the voltage lab.

- Mismatch J measures how well the model fits the readings.
- The simulation lets us measure how wrong the estimate is (state error).

Source: Classroom_Labs.ipynb, L8; cell classroom-023; four-bus DC fixture.

L8: compare two misleading-data patterns

Colab / L8

Colab L8 / run the whole cell

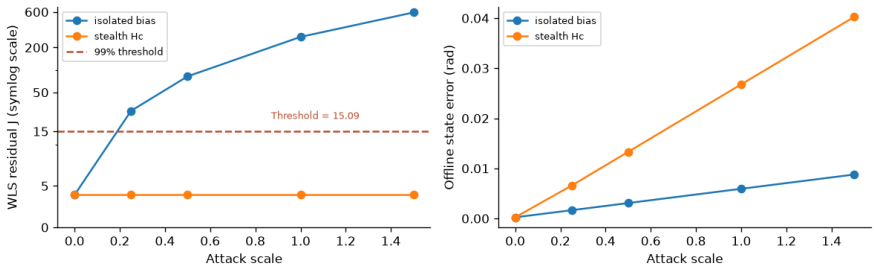
```
ATTACK_SCALES = (0., .25, .5, 1., 1.5)
started = time.perf_counter()
cyber = lab.cyber_sweep(ATTACK_SCALES)
```

- Run L8 with the default settings.
- Compare one biased reading with a coordinated change across several readings.

Source: Classroom_Labs.ipynb, L8; cell classroom-023; four-bus DC fixture.

Read the alarm and the estimation error

Colab / L8



Reference L8 plots: model fit and estimation error are different checks.

- Left: mismatch and its alarm threshold. Right: error against the simulation's known truth.
- A quiet alarm can appear alongside a growing error.

Source: Classroom_Labs.ipynb, L8; cell classroom-023; four-bus DC fixture.

A coordinated change can escape the alarm

Colab / L8

One biased reading

$J = 277.50$

Alarm

Coordinated change

$J = 3.92$

No alarm

State error: 0.02678 rad

The estimate is still wrong

L8 reference values at attack scale 1. Simulation truth supplies the state error.

- At scale 1, one biased reading gives mismatch J=277.50 and an alarm.
- A coordinated change gives J=3.92, no alarm, and 0.02678 rad of state error.

Source: Classroom_Labs.ipynb, L8; cell classroom-023; four-bus DC fixture.

Exercise 3: test a small bias

Try it in Colab / L8

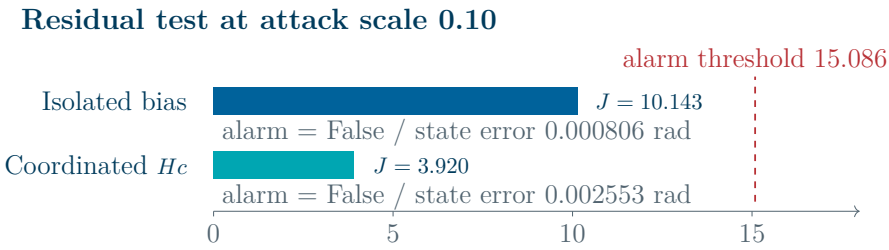
```
ATTACK_SCALES = (0., .10, .25, .5, 1., 1.5)
```

- Add .10 to ATTACK_SCALES. Predict, then rerun L8.
- At scale 0.10, compare mismatch J, the alarm and state error.
- Does no alarm prove that the readings are correct?

Source: Classroom_Labs.ipynb, L8; cell classroom-023; four-bus DC fixture.

Exercise 3 answer: no alarm can hide an error

Worked answer / L8



Fresh L8 exercise. The error uses truth known inside the simulation.

- Both alarms are False at scale 0.10, but both estimates still have error.
- No alarm means the mismatch is below this threshold. It does not establish data trust.

Source: Classroom_Labs.ipynb, L8; cell classroom-023; fresh four-bus DC residual check at scale 0.10.

A small bias can stay below the threshold

Colab / L8

$$10.14 < 15.09$$

- At scale 0.10, the mismatch is 10.14, below the 15.09 threshold.
- The operator still needs other evidence about the readings.

Source: Classroom_Labs.ipynb, L8; cell classroom-023; four-bus DC fixture.

L8: reset before the recovery example

Colab / L8

Colab L8 / run the whole cell

```
ATTACK_SCALES = (0., .25, .5, 1., 1.5)
started = time.perf_counter()
cyber = lab.cyber_sweep(ATTACK_SCALES)
```

- Restore the original attack scales and rerun L8.
- L9 uses a separate fixed recovery case.

Source: Classroom_Labs.ipynb, L8; cell classroom-023; four-bus DC fixture.

L9: ask whether a correction can be proposed

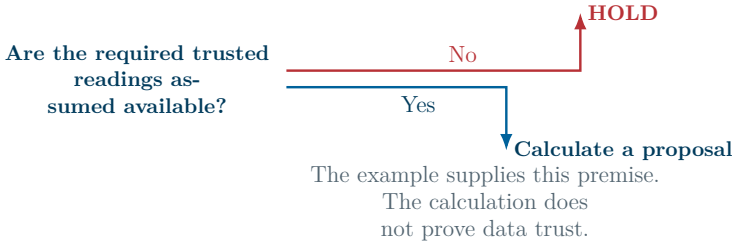
Colab / L9

- Compare one case with and without readings assumed to be trusted.
- Changing your L8 sweep does not change this fixed case.

Source: Classroom_Labs.ipynb, L9; cell classroom-026; fixed trusted-recovery fixture.

Trusted readings must come from somewhere

Colab / L9



The extra readings are assumed trustworthy for this experiment.

- This example assumes extra trustworthy readings are available.
- Neither the fitting tool nor Llama proves that these readings are authentic.

Source: Classroom_Labs.ipynb, L9; cell classroom-026; fixed trusted-recovery fixture.

L9: read HOLD or PROPOSE first

Colab / L9

Colab L9 / run the whole cell

```
without_trust = lab.trusted_response(cyber, protected_available=
    False)
with_trust = lab.trusted_response(cyber, protected_available=True)
```

- Run L9 and read the status before the proposed adjustment.
- Then check the model's returned limits and remaining assumptions.

Source: Classroom_Labs.ipynb, L9; cell classroom-026; fixed trusted-recovery fixture.

Missing trust means HOLD

Colab / L9

Executed notebook output

L9 · When can we propose a correction?

classroom-026 · output 0 (zero-based) · recorded execution 11

	condition	status	u_pu	offline_actual_violations
0	No trusted replacement channels	HOLD	NaN	NaN
1	Three trusted channels assumed	PROPOSE	0.121694	0.0

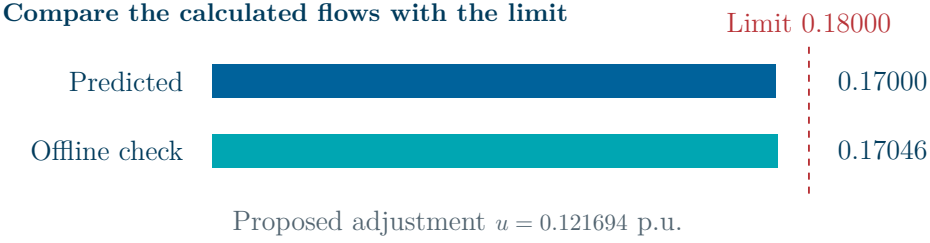
Reference L9 output for the fixed recovery example.

- Without trusted replacement readings: HOLD.
- With the trust assumption: PROPOSE a 0.121694 p.u. adjustment; the offline model checks pass.

Source: Classroom_Labs.ipynb, L9; cell classroom-026; fixed trusted-recovery fixture.

The tool calculates a small correction

Colab / L9



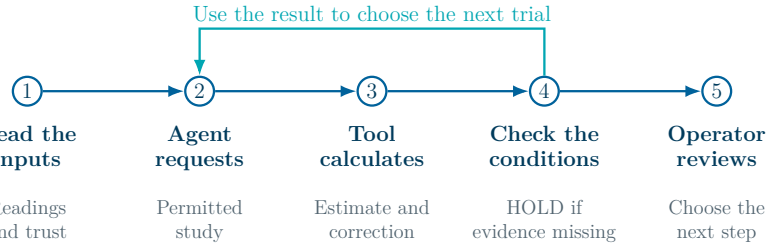
Per unit (p.u.) scales a value by a model base. These flows use the same base.

- The solver finds the smallest allowed adjustment that meets the checked branch limits.
- Predicted flow: 0.17000 p.u. Offline check: 0.17046 p.u. Limit: 0.18000 p.u.

Source: Classroom_Labs.ipynb, L9; cell classroom-026; fixed trusted-recovery fixture.

How the agent helps with suspicious readings

How an agent can help



Concept map. L9 calculates; L10B replays saved Llama requests and returned tool evidence.

- The agent can request the checks and keep missing evidence visible.
- The operator reviews trust, model coverage and the proposed correction.

Source: Classroom_Labs.ipynb, L9; cell classroom-026; fixed trusted-recovery fixture.

Recorded example: request the recovery tool

Saved cyber example / L10B preview

RECORDED REPLAY · llama3.1:8b · cyber · call 7

Input facts	The structured task already authorizes WLS_TRUSTED_REDISPATCH. Trusted sensors are explicitly assumed; no live utility measurements are supplied.
Llama choice	`WLS_TRUSTED_REDISPATCH`
Exact public reason	“This tool is authorized for the given use case and scope, and it is designed for a specific DC redispatch example with trusted measurements.”
Recorded result	The numerical tool proposes u = 0.121693667 p.u.; predicted flow12 = 0.170000 p.u. Offline synthetic-truth flow12 = 0.170461 p.u.; rating = 0.180000 p.u.

- Llama requests the supplied recovery tool.
- Python calculates a correction using the assumed trusted readings.

Source: Classroom Labs L10B: cyber replay, recorded call 7; full record in source guide.

Recorded example: keep human review

Saved cyber example / L10B preview

RECORDED REPLAY · llama3.1:8b · cyber · call 8	
Input facts	The model receives the actual numerical summary and passed_modeled_checks = True. No communication authentication, AC feeder or operating-permission check is included.
Llama choice	`REVIEW`
Exact public reason	“The provided JSON output indicates that the numerical workflow has completed successfully, and the results are within the expected scope. The 'passed_modeled_checks' field is set to True, indicating that the modeled checks have been passed. Therefore, the choice is to REVIEW the results to ensure they are accurate and within the expected range.”
Recorded result	Numerical verification: PASS. Language choice: REVIEW. Overall result: REVIEW. Human acknowledgment: PENDING; dispatch authorized = False; equipment operated = False.

- The model checks pass; human review is still pending.
- Data trust and operating permission still need review.

Source: Classroom Labs L10B: cyber replay, recorded call 8; full record in source guide.

L10: inspect the saved sequence of jobs

Colab / L10

- Run L10 to see the recorded choices, calculations and checks.
- The next diagrams give these jobs simple names.

Source: Classroom_Labs.ipynb, L10–L11; cells classroom-028 and classroom-030.

L10B: replay each saved agent example

Colab / L10B

Colab L10B / run the whole cell

```
SCENARIO = "voltage"  # Change to "restoration" or "cyber", then
   rerun this cell.
```

- Set SCENARIO to voltage, restoration or cyber, then rerun this cell.
- Read the model's request and the tool's response. This is a saved replay.

Source: Classroom_Labs.ipynb, L10B; cell classroom-llama-walkthrough; src/llama_walkthrough.py.

L12: practice checking the recommendation ID

Colab / L12

Colab L12 / run the whole cell

```
wrong_id = lab.acknowledge(mv_scope, 'wrong-id', 'Student operator
    ')
readback = lab.acknowledge(mv_scope, mv_scope['recommendation_id
    '], 'Student operator')
```

- Leave L11 disabled, then run L12.
- Compare the wrong ID with the matching ID. Equipment control stays off.

Source: Classroom_Labs.ipynb, L12; cell classroom-033; simulated readback only.

Approval must match the checked study

Colab / L12

Executed notebook output

L12 · Match the operator readback

classroom-033 · output 0 (zero-based) · recorded execution 15

Contiguous columns: state through equipment_dispatch; original row order and row index.

	state	evidence_ matches	engineering_ pass	scope_ matches	scope	acknowled ged	equipment_ dispatch
0	HOLD	True	True	False	MV phases only	False	False
1	READY_FOR_ READBACK	True	True	True	MV phases only	True	False

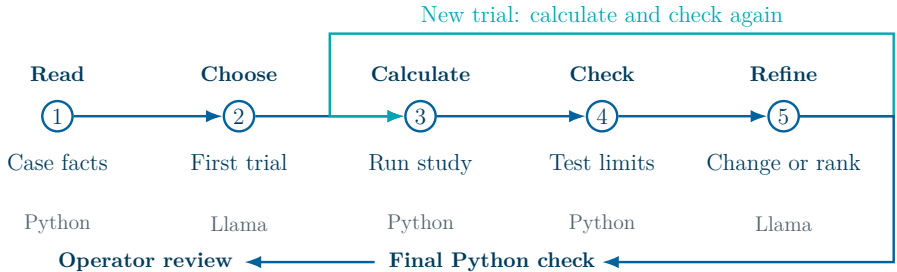
Reference L12 output. Readback is simulated.

- The wider-grid request returns HOLD. The limited teaching request permits a simulated readback.
- A wrong ID fails. This practice sends no equipment commands.

Source: Classroom_Labs.ipynb, L12; cell classroom-033; simulated readback only.

Five roles share the work

Saved five-role workflow / L10



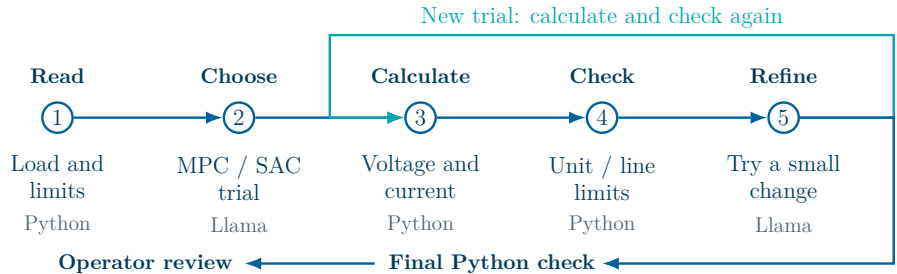
Saved five-role Python workflow; one Llama model fills two jobs.

- Python reads the case, calculates and checks. Llama chooses and refines trials.
- These are five jobs; one Llama model fills two of them.

Source: Five-role implementation: src/proposal_workflow.py; recorded results: proposal_complete_workflow.json.

Voltage: request, calculate, check, improve

Saved five-role workflow / L10



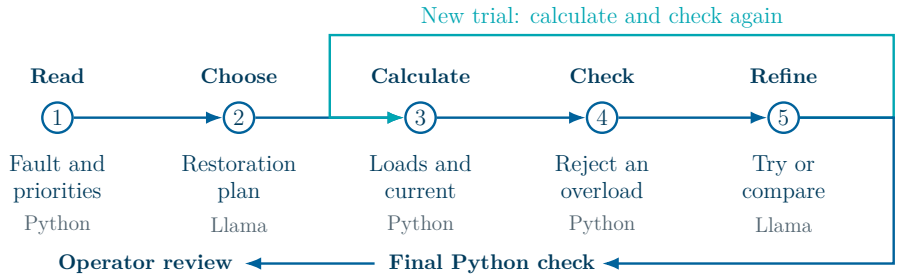
Saved L10 case: 50.75% of allowed current. Medium-voltage model only.

- Llama requests small changes to the solar settings.
- Python recalculates each trial and checks its limits before comparison.

Source: Five-role implementation: src/proposal_workflow.py; recorded results: proposal_complete_workflow.json.

Restoration: keep priorities and reject failures

Saved five-role workflow / L10



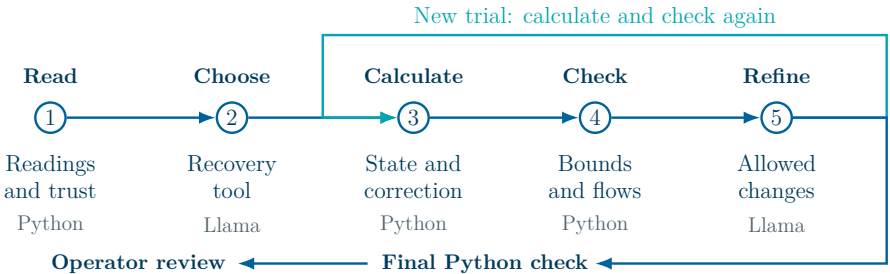
Saved L10 case: 1,843.61 kW total / 474.04 kW critical / 91.87% of allowed current.

- The goal stays fixed when new plans are tried.
- Python excludes failed trials and corrects a wrong cost order.

Source: Five-role implementation: src/proposal_workflow.py; recorded results: proposal_complete_workflow.json.

Cyber: keep the trust assumption visible

Saved five-role workflow / L10



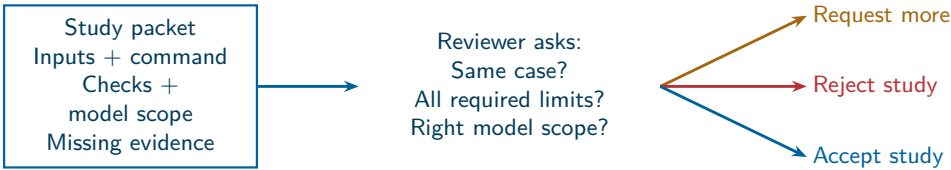
Saved L10 cyber case. Trusted measurements are an explicit assumption.

- The agent can request another permitted study or report that evidence is missing.
- It cannot make a data source trustworthy by describing it confidently.

Source: Five-role implementation: src/proposal_workflow.py; recorded results: proposal_complete_workflow.json.

The operator chooses the next step

Proposed advisory pilot / teaching design



Accepted study ≠ permission to operate equipment.

Proposed advisory pilot — teaching design

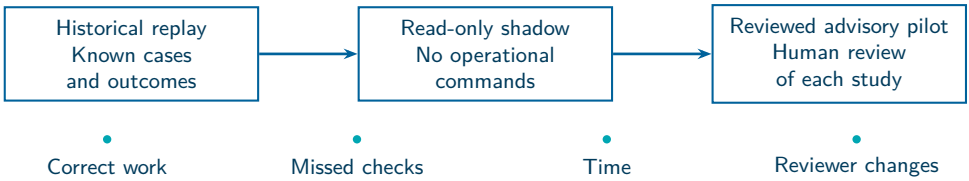
Proposed control-room use. Study review does not operate equipment.

- Review the case, tool results, checked limits and missing work.
- Request another study, reject the proposal or take the next authorized step.

Source: Teaching proposal informed by NIST OT guidance and the cited notebook checks.

Measure whether the agent helps

Proposed advisory pilot / teaching design



Compare with a fixed script + the same tools. The owner can stop the pilot.

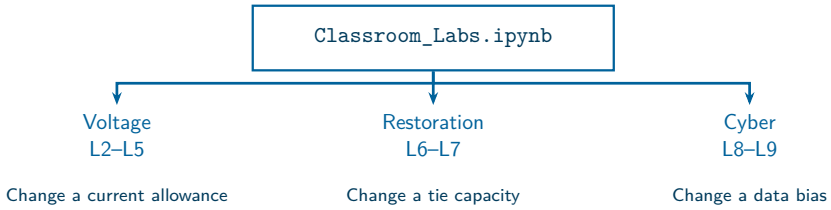
Proposed advisory pilot — teaching design

L10 comparison: same tools and trial budgets. No general performance gain is established.

- Compare it with a fixed script using the same tools and cases.
- In L10, five final costs tied; one Llama route was worse. Benefit must be measured.

Source: Teaching proposal informed by NIST OT guidance and the cited notebook checks.

Try one change and explain the result



L10B replays Llama choices in the same file. Each case uses a separate teaching network.

Keep the saved replies separate from your new experiment.

Open the three cases in Colab

- Choose one case. Change one input, rerun and save the result.
- Explain the problem, agent choice, tool result and operator decision.

Source: Setup: COLAB_GUIDE.md; Classroom_Labs.ipynb.